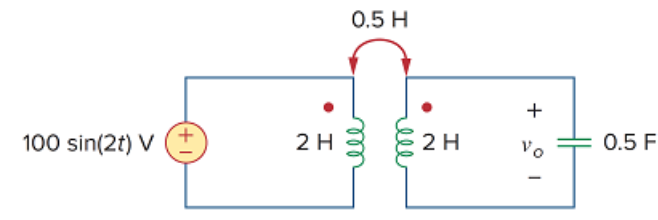


Problem

Find v_o in the circuit of Fig. 13.79.

Figure 13.79



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 13.79 in the textbook.

The voltage source in polar form is,

$$\begin{aligned} 100 \sin(2t) &= 100 \cos(2t - 90^\circ) \\ &= 100 \angle -90^\circ \text{ V} \end{aligned}$$

Calculate the impedance value of 2 H inductor.

$$\begin{aligned} X_{2H} &= j\omega L \\ &= j(2)(2) \\ &= j4 \, \Omega \end{aligned}$$

Calculate the impedance value of 0.5 H inductor.

$$\begin{aligned} X_{0.5H} &= j\omega L \\ &= j(2)(0.5) \\ &= j \, \Omega \end{aligned}$$

Calculate the impedance value of 0.5 F capacitor.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)(0.5)} \\ &= -j \, \Omega \end{aligned}$$

Step 2 of 5

Draw the frequency domain circuit.

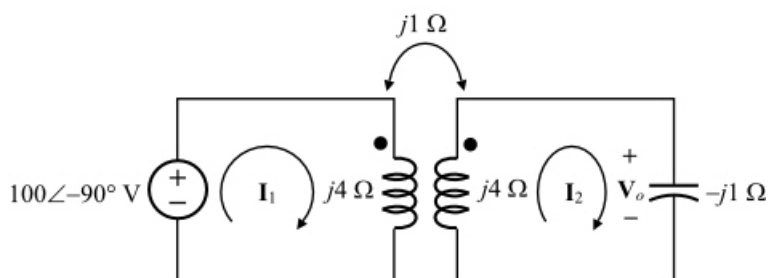


Figure 1

Step 3 of 5

Apply Kirchhoff's voltage law to loop 1.

$$-100\angle -90^\circ + j4\mathbf{I}_1 - j\mathbf{I}_2 = 0$$

$$j4\mathbf{I}_1 - j\mathbf{I}_2 = 100\angle -90^\circ \dots\dots (1)$$

Apply Kirchhoff's voltage law to loop 2.

$$j4\mathbf{I}_2 - j\mathbf{I}_1 - j\mathbf{I}_2 = 0$$

$$j3\mathbf{I}_2 - j\mathbf{I}_1 = 0$$

$$j\mathbf{I}_1 = j3\mathbf{I}_2$$

$$\mathbf{I}_1 = 3\mathbf{I}_2 \dots\dots (2)$$

Step 4 of 5

Recall equation (1).

$$j4\mathbf{I}_1 - j\mathbf{I}_2 = 100\angle -90^\circ$$

Substitute $3\mathbf{I}_2$ for $\mathbf{I}_1$ in the equation.

$$j4(3\mathbf{I}_2) - j\mathbf{I}_2 = 100\angle -90^\circ$$

$$j12\mathbf{I}_2 - j\mathbf{I}_2 = 100\angle -90^\circ$$

$$j11\mathbf{I}_2 = 100\angle -90^\circ$$

$$\begin{aligned}\mathbf{I}_2 &= \frac{100\angle -90^\circ}{j11} \\ &= 9.09\angle -180^\circ \text{ A}\end{aligned}$$

Step 5 of 5

Calculate the output voltage.

$$\mathbf{V}_o = -j1\mathbf{I}_2$$

Substitute $9.09\angle -180^\circ \text{ A}$ for $\mathbf{I}_2$ in the equation.

$$\begin{aligned}\mathbf{V}_o &= (-j1\ \Omega)(9.09\angle -180^\circ \text{ A}) \\ &= (1\angle -90^\circ \Omega)(9.09\angle -180^\circ \text{ A}) \\ &= 9.09\angle -270^\circ \text{ V}\end{aligned}$$

The output voltage in time domain is,

$$v_o(t) = 9.09\cos(2t - 270^\circ) \text{ V}$$

Therefore, the output voltage, $v_o(t)$ is $\boxed{9.09\cos(2t - 270^\circ) \text{ V}}$.